to choose the best clustering (more about this in the latest section of the chapter).

Listing 12.1 shows the code for the main method performing k-means clustering; we broke down the main steps into separate functions, to get cleaner, more easily maintainable code, and we'll delve into each step in the next pages. You can also check out implementations of the method on the book's repo on GitHub.

The algorithm can be seen as a search heuristic converging to a (local) optimum, with *slightly-more-complex-than-normal* functions to compute the score and the gradient step. At line #6, we have a stop condition that checks if the algorithm converged: if the classification hasn't changed in the last step, then the centroids will be the same as in the previous step, so any further iteration will be futile. Since these functions are quite expensive to compute at each step, and convergence is anyway **not** guaranteed, we add another stop condition by capping the execution to a maximum number of iterations.

Listing 12.2 randomCentroidInit

```
function randomCentroidInit(points, numCentroids)
                  #1
  centroids ← sample(points, numCentroids)
                           #2
  for i in {0, .., numCentroids-1}
do                                          #3
    for j in {0, .., dim-1}
do                                                  #4
      centroids[i][j] ← centroids[i][j] +
randomNoise()                          #5
```